

Deep Reinforcement Learning for Traffic Light Control in Jönköping

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Abstract—In modern smart cities, traditional fixed-time traffic signal controllers are increasingly unequipped to handle dynamic urban congestion, leading to substantial economic time loss and environmental degradation. This paper presents a comparative empirical evaluation of Deep Reinforcement Learning (DRL) frameworks applied to adaptive Traffic Signal Control (TSC) within a microscopic Simulation of Urban MObility (SUMO) environment modeled after the city center of Jönköping, Sweden. We investigate a network including of 23 traffic lights and complex roundabout-to-intersection junctions under simulated rush-hour demand profiles. A value-based Deep Q-Network (DQN) framework and an on-policy Proximal Policy Optimization (PPO) framework were deployed using a multi-agent structure with homogeneous parameter sharing across all controllers. The agents were trained using a curriculum learning strategy to optimize a queue-minimization reward function. The empirical results across ten random seeds demonstrate that both learning agents significantly outperform non-adaptive baselines. The deterministic evaluation policy of the DQN agent achieved the highest local optimizations, yielding an 87.2% reduction in vehicle waiting times, an 82.8% reduction in stops per vehicle, and a 30.9% increase in average trip velocities. However, our analysis highlighted a critical structural trade-off: by aggressively minimizing localized queues along major roads, the independent agents introduced a boundary gating effect that restricted minor incoming roads, resulting in an 11.0% deficit in total completed trips during peak demand. These findings suggest that while independent multi-agent DRL with parameter sharing offers massive corridor clearing capabilities, balancing network throughput requires future transition toward reward balancing or fully cooperative, communication-based Multi-Agent Reinforcement Learning frameworks.

I. INTRODUCTION

Rapid global urbanization and rising populations create a demand for traffic throughput optimization in order to save time, money, and resources, while increasing road safety. Traditional Traffic Signal Control (TSC) systems remain inefficient, relying on static, pre-programmed algorithms that fail to adapt to real-time traffic fluctuations. At the same time, artificial intelligence is experiencing widespread integration across decision-making domains and allows for adaptive, intelligent control systems that offer a direct path toward mitigating urban gridlock.

This paper focuses on the city center of Jönköping, Sweden, as a case study for algorithmic intersection optimization. Jönköping is chosen as its road network experiences severe bottlenecks during peak rush hours, primarily driven by

roundabouts feeding directly into closely spaced intersections controlled by traffic lights. Because these existing controllers are based on non-adaptive phase-switching strategies, they are not able to mitigate the compounding queue lengths that propagate across the interconnected junctions. Implementing a Deep Reinforcement Learning (DRL) framework provides an empirical evaluation of how Deep Learning can impact multi-intersection traffic throughput.

II. BACKGROUND

A. SUMO

Simulation of Urban MObility (SUMO) is an open-source traffic simulation suite designed to handle large-scale road networks [1]. The platform allows for modeling of streets, vehicles, and traffic signal logics. It incorporates the Traffic Control Interface (TraCI), an API that enables live state retrieval and manipulation of traffic light phases [2]. SUMO also offers fine-grained data logging capabilities for performance metrics, making it highly relevant for intelligent transportation experiments.

B. SUMO-RL

SUMO-RL is a specialized Python library designed to instantiate reinforcement learning environments using the SUMO simulator [3]. It wraps the raw simulation variables retrieved from TraCI API into a standardized interface and provides default function for observation spaces including lane density and queue lengths, and reward functions.

C. Stable Baselines3

Stable Baselines3 offers deep reinforcement learning algorithm implementations built upon the PyTorch framework [4]. It provides optimized architectures for both value-based and policy-gradient methods, including Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO), eliminating implementation bias and serving as a basis for comparative agent analysis.

III. RELATED WORKS

Zhao et al. [5] provide a comprehensive survey of DRL approaches for traffic signal control. Their review categorizes

existing methodologies based on algorithm types, agent architectures, and application scenarios. A key finding of their survey is that while single-agent algorithms perform well in isolated intersections, scaling these solutions to multi-intersection networks remains challenging. Consequently, multi-agent reinforcement learning (MARL) with parameter sharing has emerged as a practical solution for large-scale environments. Furthermore, the survey highlights the importance of evaluating these algorithms on real-world networks rather than synthetic grids. Specifically, they note the increasing use of the Monaco SUMO Traffic (MoST) scenario as a high-fidelity testbed for benchmarking decentralized MARL algorithms in realistic urban settings. In addition to Monaco, the survey emphasizes studies utilizing a real-world road network in the Xiasha District of Hangzhou, which features 48 intersections, as another critical benchmark for evaluating network-level coordination.

Recent studies have shown that Deep Reinforcement Learning (DRL) can significantly improve adaptive traffic signal control compared to traditional fixed-time approaches. Kumar et al. [6] proposed a DRL-based adaptive traffic light control system that dynamically adjusts traffic signals according to real-time traffic conditions and vehicle classifications. Their model assigns different weights to vehicles based on type, where heavy vehicles receive higher priority than light vehicles. The proposed approach employs a Deep Double Q-Learning (DDQN) algorithm with experience replay and target networks to enhance learning stability. The system was implemented using the SUMO traffic simulator and Traffic Control Interface (TraCI) for environment interaction. Experimental results demonstrated a substantial reduction in average vehicle waiting time, reaching up to 91.7% compared to conventional traffic signal control methods. However, the model mainly focuses on vehicle-weight-based prioritization and requires extensive training for optimal performance. Inspired by this work, the proposed system investigates reinforcement learning techniques for adaptive traffic signal optimization with the objective of minimizing traffic congestion and improving intersection efficiency.

IV. ENVIRONMENT

A. Network Generation

The urban road network was modeled following standard traffic scenario generation workflows [7]. First, the raw geographical data for Sweden was obtained from Geofabrik¹, and a custom bounding polygon was created using the GeoJSON² web interface to isolate the target study area³ shown in figure 1.

These boundaries were selected specifically because they contain the main roads in the Jönköping city center, including a total of five roundabouts. A challenging bottleneck within this area occurs along Östra Strandgatan (visible in the lower-right

¹<https://download.geofabrik.de/europe/sweden.html>

²<https://geojson.io/>

³<https://github.com/benbekir/traffic-rl/blob/main/simulations/networks/jkpg/artifacts/jkpg.geojson>



Fig. 1. The GeoJSON boundary polygon (shaded in grey) defining the geographic scope of the Jönköping road network.

section of the filter), where two major roundabouts operate close together and feed directly into intersections controlled by traffic signals.

To construct the network, the command-line tools *osmium* and *osmfilter* were used to clip the Swedish map data down to the defined GeoJSON polygon. Next, irrelevant infrastructure was filtered out using OpenStreetMap highway attributes, including:

- footway
- path
- pedestrian
- steps
- cycleway
- bridleway

Finally, the compiled map data was processed using the SUMO *netconvert* utility to generate the executable simulation network. The SUMO network is visualized in figure 2.

B. Simulation Scope and Limitations

The generated environment captures real-world lane geometries, intersection configurations, speed limits, and traffic light locations. However, certain simplifications were enforced to bound the complexity of the experiment. First, the simulation models passenger vehicle traffic exclusively and disregards pedestrians. Consequently, pedestrian-actuated crossing signals operate on semi-realistic default patterns rather than dynamic activation. Second, urban bus transport, which contributes heavily to public transit volume in central Jönköping, was excluded from the routing profile. We acknowledge that these abstractions deviate from real-world conditions, and the

resources/net.png

Fig. 2. The finalized road network displayed within the SUMO Netedit interface, highlighting the target traffic light placements.

simulation should be interpreted as a comparative baseline for algorithmic performance rather than an exact replica of local traffic behavior.

C. Traffic Demand Modeling

Due to the absence of publicly available traffic assignment zones or quantitative origin-destination flow data for Jönköping, a deterministic, observation-driven demand model was engineered. Relying on entirely randomized vehicle insertion across the network is unrepresentative of real traffic behavior, as it distributes vehicle density uniformly across minor residential streets and major roads alike.

Therefore, to ensure a realistic flow, an origin-destination matrix was manually constructed based on our own observations. This matrix maps commuter behaviors by defining vehicle volumes, timestamps, and network entry and exit edges. To maintain realistic routing, traffic flows are restricted exclusively to boundary edges of the network. The SUMO *duarouter* utility is then used to build the demand file and randomize vehicle routes by computing the shortest-path between the specified points and adding minor variations.

To simulate high-pressure gridlock conditions, the 8,400-second simulation timeline incorporates two distinct peak rush-hours:

- **Baseline Phase (Seconds 0–2000):** Operates under standard, off-peak baseline volumes.
- **Morning Peak (Seconds 2000–5000):** Simulates morning commuting patterns, where the majority of vehicles travel from east to west.

- **Evening Peak (Seconds 5000–8400):** Simulates evening commuting patterns, where the majority of vehicles travel from west to east.

D. Markov Decision Process Formulation

To facilitate reinforcement learning, the traffic signal control problem is formalized as a Markov Decision Process (MDP) consisting of a state observation space, a discrete action space, and a scalar reward function mapped to each agent.

1) *Observation Space:* The observation vector for each traffic signal agent provides intersection’s operational state. The state is represented as a multi-dimensional vector structured as follows [8]:

$$O = [P_{\text{one-hot}}, M_{\text{green}}, D_1, \dots, D_n, Q_1, \dots, Q_n] \quad (1)$$

Where:

- $P_{\text{one-hot}}$: A one-hot encoded vector representing the currently active green phase configuration.
- M_{green} : An indicator specifying whether the minimum green phase duration has elapsed.
- D_i : The normalized vehicle density of incoming lane i , calculated by dividing the current number of present vehicles by the maximum physical vehicle capacity of that lane.
- Q_i : The normalized queue length of incoming lane i , defined as the number of halted vehicles (traveling at a velocity below 0.1 m/s) divided by the total lane capacity.

2) *Action Space:* The action space is modeled as a discrete set where each action corresponds to a specific green phase configuration at the intersection [9]. Every 5 seconds, the agent evaluates the state observation and selects the next phase configuration. To maintain traffic safety and mimic physical infrastructure constraints, any action that triggers a change in the active phase configuration is automatically prepended by a yellow phase lasting a fixed 2 seconds.

3) *Reward Function:* While the underlying SUMO-RL framework supports various native reward formulations, including changes in cumulative vehicle delay, average velocity metrics, pressure optimization, and carbon dioxide emissions, this study implements the specialized “queue” reward function [10]. The reward is mathematically defined as the negative sum of all queued vehicles across all incoming lanes feeding into the intersection agent [11]:

$$R_t = - \sum_{i=1}^n \text{HaltingVehicles}_i(t) \quad (2)$$

V. EXPERIMENT SETUP

A. Reinforcement Learning Agents

To control the target Jönköping network, the problem is structured as an Independent Multi-Agent Reinforcement Learning (I-MARL) framework comprising 23 distinct traffic light agents. To ensure computational efficiency and rapid convergence across a network of this scale, a homogeneous parameter-sharing architecture was implemented. Under this

paradigm, all 23 agents feed their experiences into a single, centralized policy network during training and pull weights from this shared network during execution. While the agents act independently and make decentralized phase choices based on their observations, parameter sharing enables them to collectively optimize a generalized traffic control policy. Within this framework, we evaluate and compare two distinct classes of DRL architectures: a value-based DQN method and a policy-gradient PPO method.

1) *Deep Q-Networks (DQN)*: Deep Q-Networks combine traditional Q-learning with deep neural networks to approximate the optimal action-value function [12]. DQN is an off-policy algorithm that stores past experiences in a large replay buffer, allowing it to sample random, uncorrelated historical batches to stabilize training. During evaluation, the agent greedily selects actions with the estimated Q-values, picking the highest scoring traffic phase configuration for a given state.

2) *Proximal Policy Optimization (PPO)*: Proximal Policy Optimization is an on-policy, actor-critic framework that directly optimizes the agent’s control policy [13]. Instead of assigning strict value scores to actions, PPO maps states directly to a probability distribution over the available traffic phases. To decrease the chance of catastrophic forgetting off destabilizing gradient steps during chaotic traffic conditions, PPO clips its policy updates within a safe bound. It learns strictly from current environment steps, discarding data after each update phase.

B. Hyperparameter Configurations and Training Protocol

Both algorithms were initialized using parameter baselines optimized over a preliminary 100-episode training phase. Using these optimized baselines, a curriculum learning strategy was deployed to progressively scale the traffic demand density across three distinct multiplier steps: 0.75, 0.875, and 1.0. For the final evaluation, the DQN agent was trained for 100 episodes per curriculum phase to maximize historical experience accumulation. However, initial trials showed that PPO’s performance degraded with the extended training horizon. To mitigate this policy degradation, the training duration for the PPO agent was shortened to 50 episodes per curriculum phase. The finalized hyperparameter configurations utilized for this execution are compiled in Table I.

TABLE I
HYPERPARAMETER CONFIGURATIONS FOR DQN AND PPO AGENTS.

Algorithm	Hyperparameter	Value
DQN	Learning Rate	1×10^{-3}
	Buffer Size	500,000
	Batch Size	128
	Exploration Fraction	0.1
	Initial Epsilon (ϵ_{init})	1.0
	Final Epsilon (ϵ_{final})	0.05
PPO	Learning Rate	1×10^{-4}
	Rollout Horizon (n_{steps})	512
	Batch Size	256
	Epochs per Update (n_{epochs})	10
	Entropy Coefficient (ent_coef)	0.2
	Clip Range (ϵ_{clip})	0.3

The DQN agent implements an ϵ -greedy exploration strategy where the exploration rate lineary decays from an initial value of 1.0 down to a final value of 0.05 across the first 10% of the total training steps. It uses an experience replay memory buffer of 500,000 steps to maintain a large historical distribution of past traffic patterns, sampling mini-batches of 128 experiences per optimization step.

Conversely, the PPO agent operates under a more conservative learning rate of 1×10^{-4} to stabilize gradient updates. The agent collects experiences over 512 steps before performing 10 optimization epochs across mini-batches of 256 steps. To preserve policy exploration, an elevated entropy coefficient of 0.2 was selected. Furthermore, a policy clipping threshold of 0.3 enforces strict bounding constraints on the policy updates, preventing catastrophic divergence when the agent is exposed to severe rush-hour gridlock configurations.

Every configuration was evaluated across 10 independent random seeds to guarantee statistical significance and capture structural performance variance.

VI. RESULTS

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Fig. 3. Multi-run traffic signal optimization results across 10 random seeds. Clockwise from top-left: (a) Cumulative completed vehicles over the 8,400-second timeline with 95% confidence intervals; (b) Probability density distribution of global vehicle time loss; (c) Average trip velocity mapped against route distance; (d) Scatter visualization contrasting average vehicle stops against mean waiting times (seconds).

The empirical results collected across 10 independent random seeds demonstrate that both DRL frameworks outperform the baseline controller across most metrics. As detailed in Table II, the Deep Q-Network (DQN) agent achieved the most significant optimizations, engineering a massive **87.2% reduction in average vehicle waiting time** and an **82.8% reduction in stops per vehicle** compared to the static baseline.

The Proximal Policy Optimization (PPO) agent also yielded substantial throughput enhancements, reducing waiting times by 64.0% and vehicular stops by 54.2%.

TABLE II
IMPROVEMENT OVER BASELINE (10-RUN AVERAGE).

Metric	DQN	PPO
Waiting time	-87.2%	-64.0%
Time loss	-77.3%	-55.2%
Stops per vehicle	-82.8%	-54.2%
Average speed	+30.9%	+20.5%
Completed trips	-11.0%	-6.8%

This performance difference is also illustrated in Figure 3. The global time loss distribution (top-right plot) highlights that the baseline traffic system exhibits a heavily skewed tail, confirming severe gridlocks across the network, while both learning agents compress this distribution. DQN tightly centers vehicle time loss around a highly efficient peak below 50 seconds. This efficiency translates into higher network velocities, as shown in the velocity-distance profile in Table III, the DQN agent maintains an average trip speed of 8.81 m/s (+30.9%), whereas PPO achieves 8.10 m/s (+20.5%), both consistently outperforming the baseline profile (6.73 m/s).

TABLE III
PERFORMANCE COMPARISON ACROSS 10 RUNS (AVERAGED).

Metric	Baseline	DQN	PPO
Completed trips	5,837	5,192	5,443
Avg. waiting time	97.4s	12.5s	35.1s
Avg. time loss	199.4s	45.2s	89.3s
Avg. stops per vehicle	6.48	1.11	2.97
Avg. speed	6.73 m/s	8.81 m/s	8.10 m/s
Total waiting time	568,322s	64,689s	201,287s
Simulation end time	8,398s	8,398s	8,157s

The superiority of DQN over PPO in this environment can be traced back to their architectural differences. Because DQN computes explicit action-value estimations and executes a deterministic *argmax* selection policy during evaluation, it enforces stable phase-clearing queues. Conversely, PPO’s on-policy framework samples actions from a probability distribution. Even during evaluation, there is variance in PPO’s policy, causing minor policy shifts that are less effective at resolving the queues. This variance is reflected in the wider 95% confidence intervals shaded along PPO’s trajectories in the top-left plot of Figure 3.

However, an important trade-off is observed regarding the total volume of completed trips. The baseline system successfully processes 5,837 vehicles, while PPO and DQN exhibit a deficit of -6.8% and -11.0% completed trips, respectively. In microscopic traffic modeling, this discrepancy indicates a classic reinforcement learning phenomenon known as *boundary gating*. Because the reward function heavily penalizes the absolute number of halting vehicles inside the visible intersection lanes ($R_t = -\sum \text{HaltingVehicles}_i$), the agents learn that the most effective way to minimize internal

intersection queues is to prolong green phases for the main roads (such as Kungsgatan).

While this strategy creates a highly fluid experience for vehicles on the main roads, it starves minor incoming lanes. Consequently, compounding queues build up outside the active simulation boundaries, causing SUMO to restrict or delay new vehicle spawning. This explains why the cumulative completed trips curve in the top-left plot shows the baseline pulling ahead during the peak evening rush hour (after second 5,000). The baseline prioritizes all lanes equally the cost of global gridlock, whereas the adaptive DRL agents prioritize network fluidity by restricting input volumes during peak directional pressure.

VII. CONCLUSION

This paper presented an empirical evaluation of DRL frameworks for adaptive TSC within a simulation of the Jönköping city center network.

Our analysis revealed that the DQN agent achieved the most deterministic and aggressive optimization, slashing vehicle waiting times by 87.2% and vehicle stops by 82.8% while significantly elevating average corridor speeds. PPO also showed improvements, though its stochastic policy induced a higher variance that proved less efficient at clearing congestion spikes.

The results highlight the trade-off between intersection efficiency and global network throughput. While the queue-minimization reward function successfully optimized vehicle velocity and minimized delays for primary traffic lanes, it also incentivized the agents to ignore minor roads, resulting in a deficit of up to 11.0% in completed network trips during severe evening rush-hour peaks.

Future work could focus on rectifying this deficiency. To prevent agents from isolating secondary roads, the reward function can be augmented with spatial metrics or queue-length thresholds that penalize extreme asymmetry across incoming lanes. Additionally, expanding the independent multi-agent setup into a fully cooperative, communication-based Multi-Agent Reinforcement Learning framework will allow adjacent intersections to actively coordinate phase changes, smoothing macro-level transit waves across Jönköping’s broader urban infrastructure.

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